

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 6

DUE: WEDNESDAY, MARCH 11, 2026, **BEFORE 23:59PM**

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - Your solutions must be *handwritten*. For instance on a tablet/iPad, or with pen and paper scanned to a pdf.
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. In this exercise we will write $\mathbb{Z}[S]$ for the free abelian group generated by a set S .

Let S_0 and S_1 be sets, let $f, g : S_1 \rightarrow S_0$ be two functions, and let $\sim$ be the equivalence relation on S_0 generated by $f(s) \sim g(s)$ for all $s \in S_1$. Then the quotient map $S_0 \rightarrow S_0/\sim$ induces a homomorphism of free abelian groups

$$\mathbb{Z}[S_0] \xrightarrow{q} \mathbb{Z}[S_0/\sim]$$

- (i) Prove that q is surjective, and that the kernel of q is generated by the subset

$$\{f(s) - g(s) \in \mathbb{Z}[S_0] \mid s \in S_1\}.$$

- (ii) Construct a map $\partial : \mathbb{Z}[S_1] \rightarrow \mathbb{Z}[S_0]$ whose image equals $\text{Ker}(q)$.

Remark (not needed for solving the exercise): In language that we shall soon introduce, this shows that

$$\mathbb{Z}[S_1] \xrightarrow{\partial} \mathbb{Z}[S_0] \xrightarrow{q} \mathbb{Z}[S_0/\sim] \rightarrow 0$$

is an “exact sequence”.

2. *This question is a continuation of the previous.*

Let X be a topological space. In the notation from the previous question, set $S_0 = \text{Sin}_0(X)$, $S_1 = \text{Sin}_1(X)$, $f = d_0$, and $g = d_1$. As in HW5, let $\pi_0(X)$ denote the set of path components of X .

- (a) Explain how to identify $S_0/\sim$ with $\pi_0(X)$, the set of path components of X .
- (b) Explain why the map q from the previous question induces an isomorphism

$$H_0(X) \rightarrow \mathbb{Z}\pi_0(X).$$

3. Let $p : \mathbb{R} \rightarrow S^1$ be the usual covering map, i.e., $p(t) = (\cos(2\pi t), \sin(2\pi t))$.

- (a) Prove that for any $\sigma \in \text{Sin}_1(S^1)$, there exists a $\mu \in \text{Sin}_1(\mathbb{R})$ such that $\sigma = p \circ \mu$.
- (b) Prove that the formula $w(\sigma) = \mu((1, 0)) - \mu((0, 1))$ gives a well defined function

$$w : \text{Sin}_1(S^1) \rightarrow \mathbb{R}.$$

- (c) Extend by linearity to a homomorphism $w : C_1(S^1) \rightarrow \mathbb{R}$. Prove that the composition

$$C_2(S^1) \xrightarrow{\partial} C_1(S^1) \xrightarrow{w} \mathbb{R}$$

is zero, and deduce that

$$H_1(S^1) \rightarrow \mathbb{R}$$

$$[c] \mapsto w(c)$$

is a well defined homomorphism.

4. *This question is a continuation of the previous.*

Let $\sigma \in \text{Sin}_1(S^1)$ be defined by the formula

$$\sigma((t_0, t_1)) = (\cos(2\pi t_1), \sin(2\pi t_1)).$$

- (a) Prove that $\partial\sigma = 0 \in C_0(S^1)$, so that $[\sigma] \in H_1(S^1)$ is well defined.
- (b) Use the function $H_1(S^1) \rightarrow \mathbb{R}$ from the previous exercise to show that $[\sigma] \neq 0 \in H_1(S^1)$.

5. Let (X, x) be a based space. We will define a category $\mathbf{Cov}_{(X, x)}$ whose objects are the triples (Y, y, p) where Y is a topological space, $y \in Y$, and $p : Y \rightarrow X$ is a covering map with $p(y) = x$. The morphisms $(Y, y, p) \rightarrow (Y', y', p')$ are the continuous maps $f : Y \rightarrow Y'$ such that $p' \circ f = p$ and $f(y) = y'$. Composition in $\mathbf{Cov}_{(X, x)}$ is defined by composition of functions. *It may be used without proof that this indeed defines a category.*

Prove that a morphism $f : (Y, y, p) \rightarrow (Y', y', p')$ is an isomorphism in $\mathbf{Cov}_{(X, x)}$ if and only if the underlying map $f : Y \rightarrow Y'$ is a homeomorphism.